北京交通大學

信号与系统

School of Electronic
and Information Engineering

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2025.3

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System Frequency Domain Analysis

- Why Perform Frequency Domain Analysis of a System
- Frequency Domain Analysis of Continuous Time LTI Systems
- Frequency Domain Analysis of Discrete Time LTI Systems
- Signal Amplitude Modulation and Demodulation
- Frequency Domain Analysis of System Using MATLAB

Why Perform Frequency Domain Analysis of a System?

Signal disturbed by noise

Signal denoising

System 1 processed (filtered) result

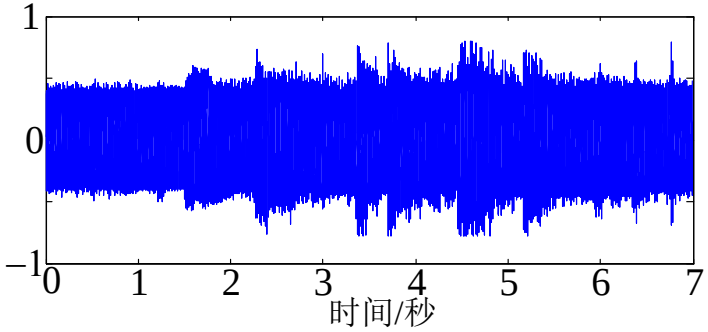
System 2 processed (filtered) results

System 3 processed (filtered) results

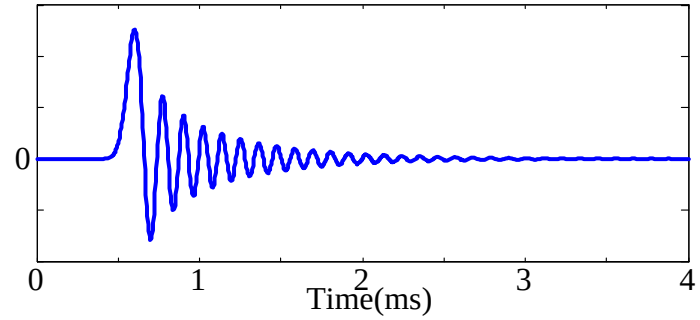
Why do different systems have different processing effects?

Why Perform Frequency Domain Analysis of a System?

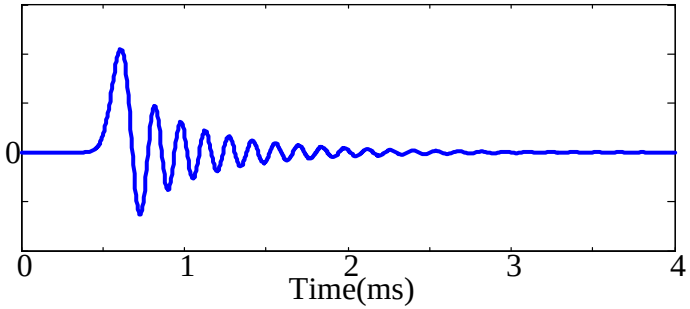
Signal denoising



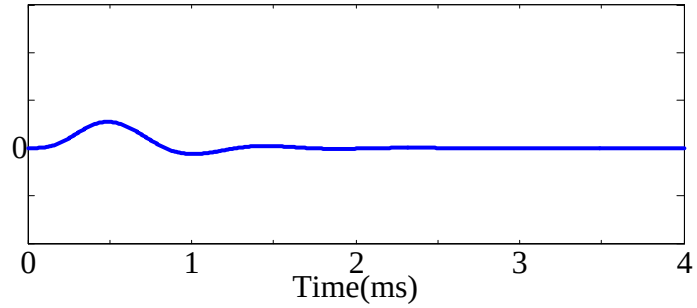
Time Domain Waveform of Interfering Signal



Impulse Response of System 1



Impulse Response of System 2

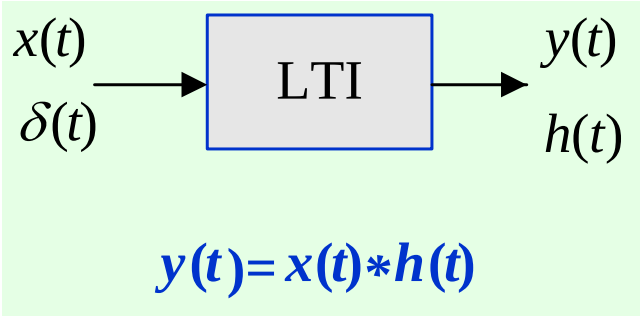


Impulse Response of System 3

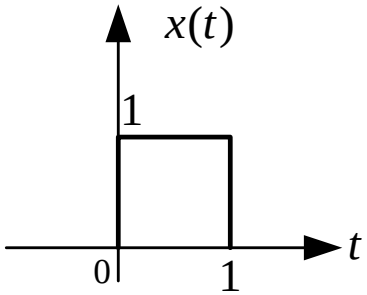
Time Domain Interpretation of Processing Results - Difficult

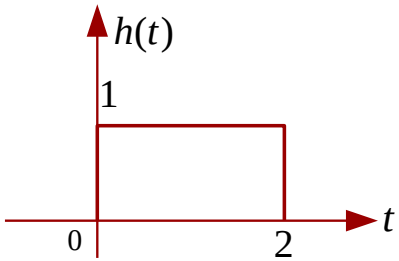
Why Perform Frequency Domain Analysis of a System?

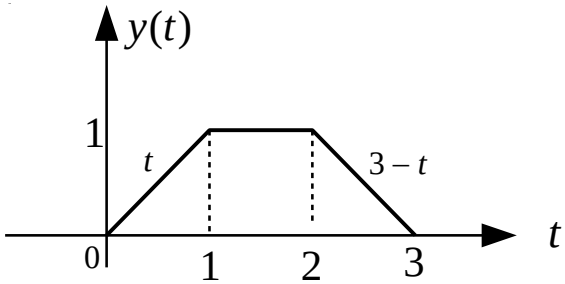
Signal transmission



$$y(t) = x(t) * h(t)$$







System Frequency Domain Analysis

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- Frequency Domain Analysis of Continuous Time LTI Systems
- Frequency Domain Analysis of Discrete Time LTI Systems
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Frequency Domain Analysis of Continuous Time LTI Systems

- ※ Frequency Domain Description of Continuous Time LTI Systems
- ※ Aperiodic signal through continuous LTI system
- ※ Periodic signal through continuous LTI system
- ※ Distortionless Transmission System
- ※ Ideal analog filter
- ※ Application example of continuous system frequency domain analysis

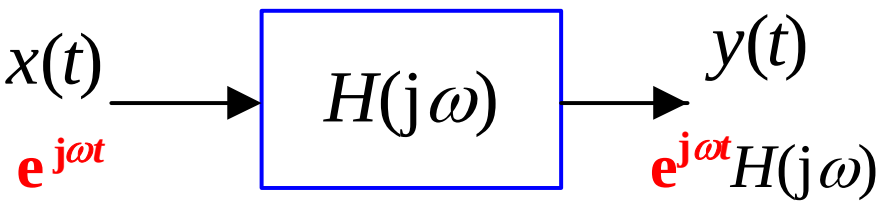
1. Frequency-domain description of continuous-time LTI systems

- Response of the imaginary exponential signal $e^{j\omega t}$ through the LTI system

$$y_{zs}(t) = e^{j\omega t} * h(t) = \int_{-\infty}^{\infty} e^{j\omega(t-\tau)} h(\tau)\mathrm{d}\tau$$

$$= e^{j\omega t} \underbrace{\int_{-\infty}^{\infty} e^{-j\omega\tau} h(\tau)\mathrm{d}\tau}_{H(j\omega)}$$

$$= e^{j\omega t} H(j\omega)$$



1. Frequency-domain description of continuous-time LTI systems

- **System Frequency Response $H(\mathrm{j}\omega)$**

$$H(\mathrm{j}\omega)=\mathscr{F}[h(t)]=\int_{-\infty}^{\infty}h(t)\mathrm{e}^{-\mathrm{j}\omega t}\mathrm{d}t$$

- **Magnitude Response vs. Phase Response**

$$H(\mathrm{j}\omega)=|H(\mathrm{j}\omega)|\,\mathrm{e}^{\mathrm{j}\varphi(\omega)}$$

magnitude response

phase response

If $h(t)$ is a real function, the amplitude response $|H(\mathrm{j}\omega)|$ is Even function of ω,

phase response $\varphi(\omega)$ is odd function of ω。

1. Frequency-domain description of continuous-time LTI systems

- **Another definition of frequency response $H(j\omega)$**

Frequency Domain Representation of Zero State Responses

$$y_{zs}(t) = x(t) * h(t)$$

Using the convolution feature of the Fourier transform

$$Y_{zs}(j\omega) = X(j\omega)H(j\omega)$$

$$H(j\omega) = \frac{Y_{zs}(j\omega)}{X(j\omega)}$$

1. Frequency-domain description of continuous-time LTI systems

- **General Form of Frequency Response $H(\mathrm{j}\omega)$ of Continuous Time LTI System**

If the differential equation describing the continuous LTI system is

$$a_n y^{(n)}(t) + a_{n-1} y^{(n-1)}(t) + \cdots + a_1 y'(t) + a_0 y(t) =$$
$$b_m x^{(m)}(t) + b_{m-1} x^{(m-1)}(t) + \cdots + b_1 x'(t) + b_0 x(t)$$

Using the Fourier transform differential characteristics, the frequency domain equation describing the system can be obtained

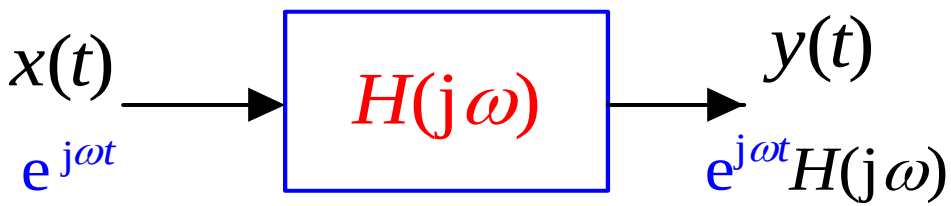
$$[a_n(\mathrm{j}\omega)^n + a_{n-1}(\mathrm{j}\omega)^{n-1} + \cdots + a_1(\mathrm{j}\omega) + a_0]Y_{zs}(\mathrm{j}\omega) =$$
$$[b_m(\mathrm{j}\omega)^m + b_{m-1}(\mathrm{j}\omega)^{m-1} + \cdots + b_1(\mathrm{j}\omega) + b_0]X(\mathrm{j}\omega)$$

From the definition of the frequency response of a continuous system, we can get

$$H(\mathrm{j}\omega) = \frac{Y_{zs}(\mathrm{j}\omega)}{X(\mathrm{j}\omega)} = \frac{b_m(\mathrm{j}\omega)^m + b_{m-1}(\mathrm{j}\omega)^{m-1} + \cdots + b_1(\mathrm{j}\omega) + b_0}{a_n(\mathrm{j}\omega)^n + a_{n-1}(\mathrm{j}\omega)^{n-1} + \cdots + a_1(\mathrm{j}\omega) + a_0}$$

1. Frequency-domain description of continuous-time LTI systems

- **The physical meaning of $H(\mathrm{j}\omega)$**



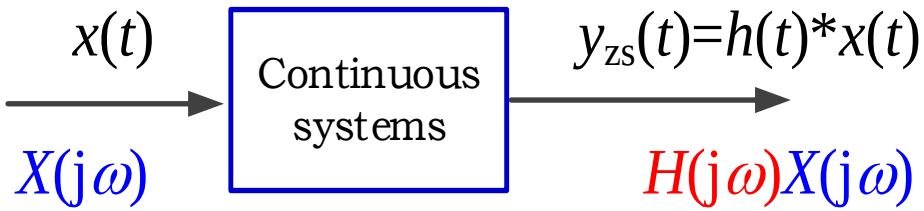
Response of the continuous LTI system through the imaginary exponential signal $e^{\mathrm{j}\omega t}$ of frequency w

It is still an imaginary exponential signal with the same frequency. The change of signal is determined by $H(\mathrm{j}\omega)$.

$H(\mathrm{j}\omega)$ reflects the transmission characteristics of the continuous system to imaginary exponential signals of different frequencies.

1. Frequency-domain description of continuous-time LTI systems

- **The physical meaning of $H(j\omega)$**



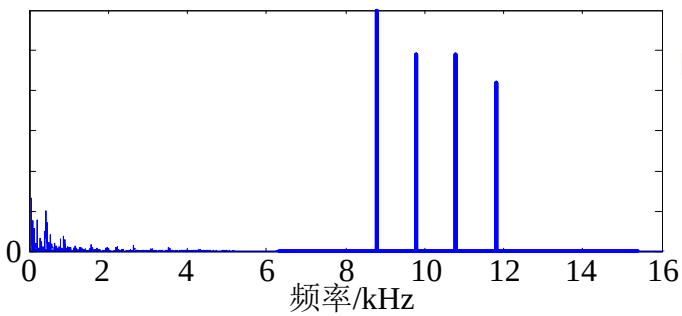
The LTI system changes the input signal spectrum $X(j\omega)$ into $H(j\omega)\,X(j\omega)$, and the change rule is completely determined by $H(j\omega)$.

$H(j\omega)$ reflects the transmission characteristics of the system to different frequency components of the input signal.

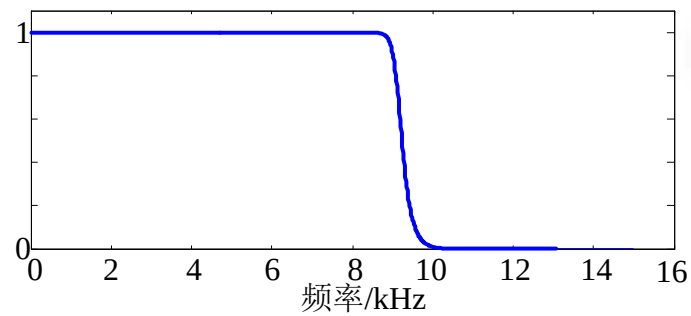
1. Frequency-domain description of continuous-time LTI systems

Signal denoising

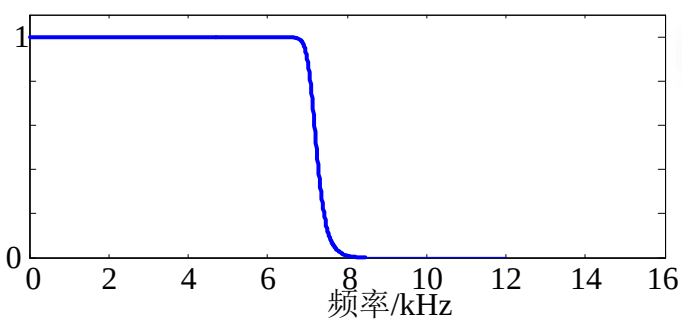
Influence of the frequency domain characteristics of the system on the filtering result



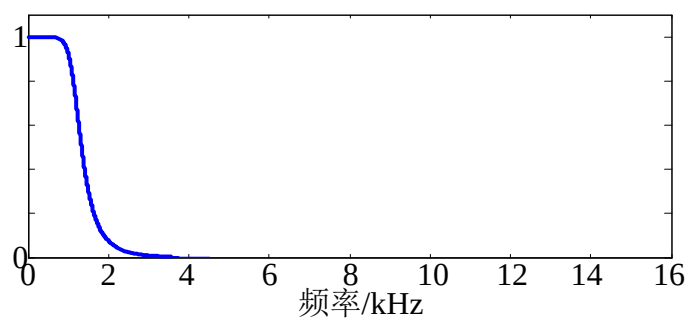
Spectrum of interfered signal



Magnitude Response of System 1



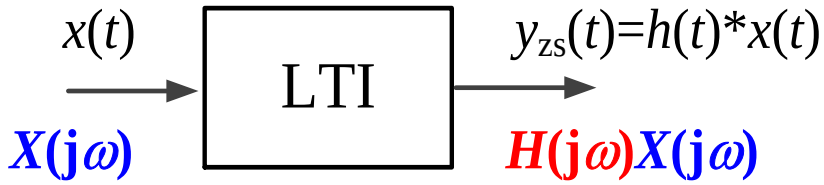
Magnitude Response of System 2

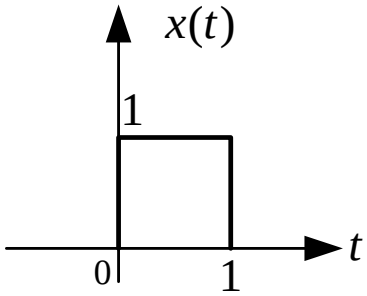


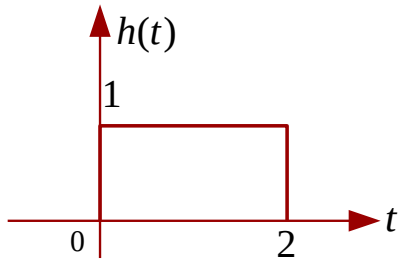
Magnitude Response of System 3

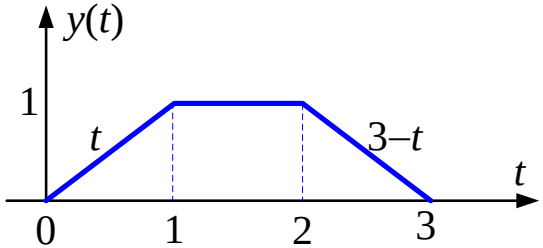
1. Frequency-domain description of continuous-time LTI systems

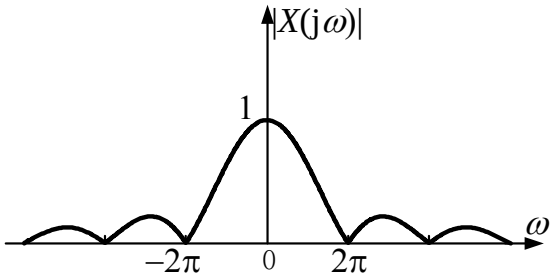
Signal transmission

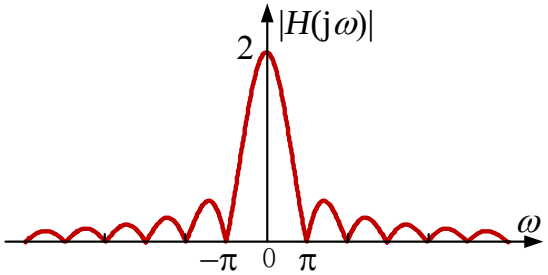


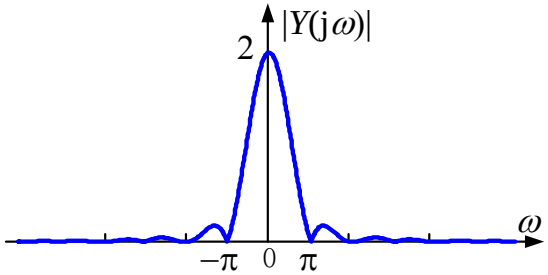












1. Frequency-domain description of continuous-time LTI systems

➢Calculation Method of Frequency Response of Continuous LTI System

$$H(\mathrm{j}\omega) = \mathscr{F}\{h(t)\} = \int_{-\infty}^{\infty} h(t)\mathrm{e}^{-\mathrm{j}\omega t}\mathrm{d}t$$

- Computed from the Fourier transform of the impulse response of the continuous LTI system;
- Computed from the differential equations describing the continuous-time LTI system;
- The calculations are performed from a frequency-domain equivalent circuit model of the actual circuit.

Calculation Method of Frequency Response of Continuous LTI System

- From the impulse response of a continuous LTI system, calculate the frequency response of the system

$$H(\mathrm{j}\omega) = \mathscr{F}\{h(t)\} = \int_{-\infty}^{\infty} h(t)\mathrm{e}^{-\mathrm{j}\omega t}\mathrm{d}t$$

Calculation Method of Frequency Response of Continuous LTI System

Ex. It is known that the impulse response of a continuous LTI system is $\boldsymbol{h(t) = (e^{-t} - e^{-2t})\, u(t)}$, Find the frequency response $H(\mathrm{j}\omega)$ of the system.

Sol' n: Using the relationship between $\boldsymbol{H(\mathrm{j}\omega)}$ and $\boldsymbol{h(t)}$

$$H(\mathrm{j}\omega) = \mathscr{F}\{h(t)\} = \frac{1}{\mathrm{j}\omega + 1} - \frac{1}{\mathrm{j}\omega + 2}$$

$$= \frac{1}{(\mathrm{j}\omega)^2 + 3(\mathrm{j}\omega) + 2}$$

Calculation Method of Frequency Response of Continuous LTI System

- Calculate the frequency response of the system from the differential equations describing the continuous LTI system.

If the differential equation describing the LTI system is

$$y'''(t)+a_2y''(t)+a_1y'(t)+a_0y(t)=b_2x''(t)+b_1x'(t)+b_0x(t)$$

Using the differential characteristics of the Fourier transform, the frequency domain expression of the differential equation is

$$[(\mathrm{j}\omega)^3+a_2(\mathrm{j}\omega)^2+a_1(\mathrm{j}\omega)+a_0]Y_{zs}(\mathrm{j}\omega)=$$

$$[b_2(\mathrm{j}\omega)^3+b_1(\mathrm{j}\omega)^2+b_0]X(\mathrm{j}\omega)$$

From the definition of the frequency response

$$H(\mathrm{j}\omega)=\frac{Y_{zs}(\mathrm{j}\omega)}{X(\mathrm{j}\omega)}=\frac{b_2(\mathrm{j}\omega)^2+b_1(\mathrm{j}\omega)+b_0}{(\mathrm{j}\omega)^3+a_2(\mathrm{j}\omega)^2+a_1(\mathrm{j}\omega)+a_0}$$

Calculation Method of Frequency Response of Continuous LTI System

Ex. It is known that the differential equation describing a continuous LTI system is

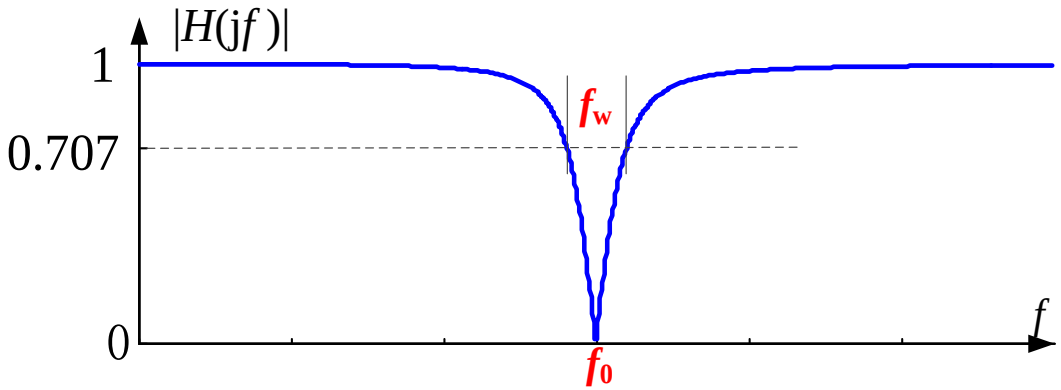
$$y''(t)+(2\pi f_{\mathrm{w}})y'(t)+(2\pi f_0)^2 y(t)=x''(t)+(2\pi f_0)^2 x(t)$$

Find the frequency response $H(\mathrm{j}\omega)$ of the system.

Solution: From the Fourier transform properties and the definition of frequency response, we can get

$$[(\mathrm{j}\omega)^2+2\pi f_{\mathrm{w}}(\mathrm{j}\omega)+(2\pi f_0)^2]Y_{\mathrm{zs}}(\mathrm{j}\omega)=[(\mathrm{j}\omega)^2+(2\pi f_0)^2]X(\mathrm{j}\omega)$$

$$H(\mathrm{j}\omega)=\frac{(\mathrm{j}\omega)^2+(2\pi f_0)^2}{(\mathrm{j}\omega)^2+(2\pi f_{\mathrm{w}})(\mathrm{j}\omega)+(2\pi f_0)^2}$$

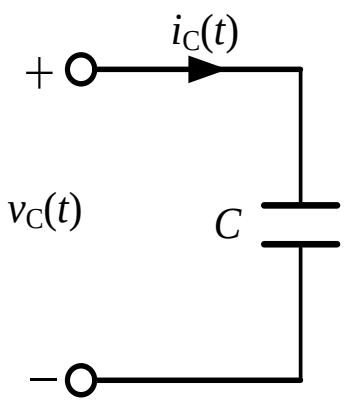


Band-stop filter with a center frequency of f_0 Hz and a 3dB bandwidth of f_{w} Hz

Calculation Method of Frequency Response of Continuous LTI System

- Calculate the system frequency response based on the frequency-domain equivalent model of the circuit

Frequency Domain Model of Capacitance



$$i_C(t) = C\frac{\mathrm{d}v_C(t)}{\mathrm{d}t}$$

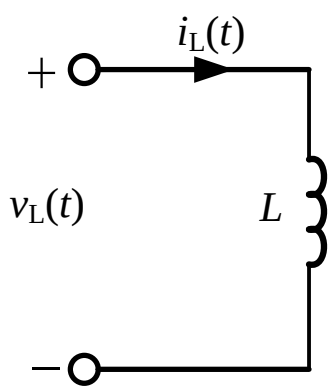
$$I_C(\mathrm{j}\omega) = C\mathrm{j}\omega V_C(\mathrm{j}\omega)$$

$$\frac{V_C(\mathrm{j}\omega)}{I_C(\mathrm{j}\omega)} = \frac{1}{\mathrm{j}\omega C}$$

Calculation Method of Frequency Response of Continuous LTI System

- Calculate the system frequency response based on the frequency-domain equivalent model of the circuit

Frequency Domain Model of Inductor



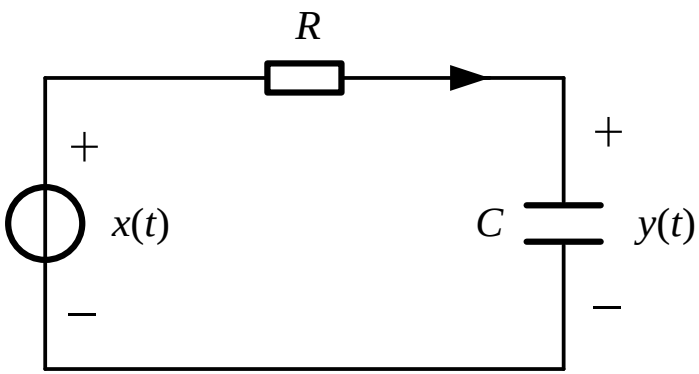
$$v_L(t) = L\frac{\mathrm{d}i_L(t)}{\mathrm{d}t}$$

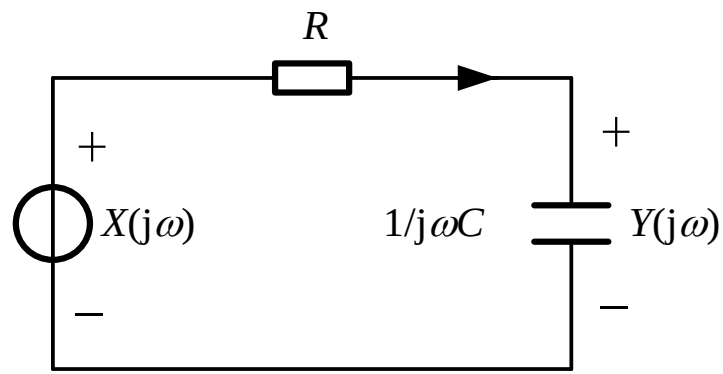
$$V_L(\mathrm{j}\omega) = L\mathrm{j}\omega I_L(\mathrm{j}\omega)$$

$$\frac{V_L(\mathrm{j}\omega)}{I_L(\mathrm{j}\omega)} = \mathrm{j}\omega L$$

Calculation Method of Frequency Response of Continuous LTI System

Example The RC circuit system shown in the figure, the excitation voltage source is $x(t)$, the output voltage $y(t)$ is the voltage $v_C(t)$ across the capacitor, and the initial state of the circuit is zero. Find the frequency response $H(\mathrm{j}\omega)$ and impulse response $h(t)$ of the system.





Solution: The frequency domain model of the RC circuit is shown in the figure,

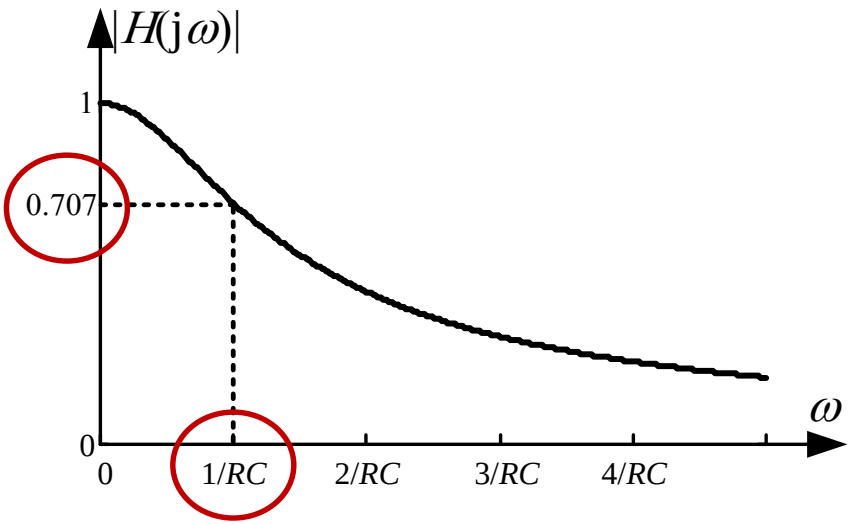
By the basic principle of the circuit there are

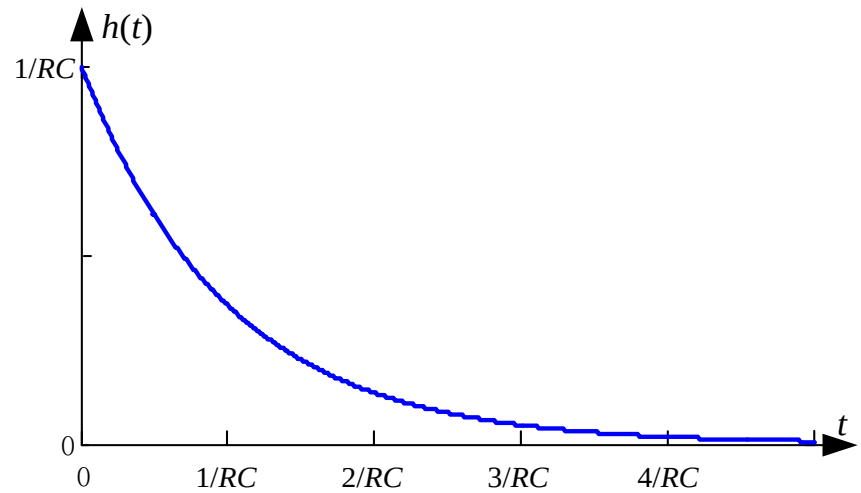
$$H(\mathrm{j}\omega)=\frac{Y(\mathrm{j}\omega)}{X(\mathrm{j}\omega)}=\frac{1/\mathrm{j}\omega C}{R+\dfrac{1}{\mathrm{j}\omega C}}=\frac{1/RC}{\mathrm{j}\omega+1/RC}$$

Calculation Method of Frequency Response of Continuous LTI System

solution: By Fourier inverse transformation, the *impulse response h(t)* of the system can be obtained as

$$h(t)=\mathscr{F}\{H(\mathrm{j}\omega)\}=\mathscr{F}\{\frac{1/RC}{\mathrm{j}\omega+1/RC}\}=\frac{1}{RC}\mathrm{e}^{-(1/RC)t}u(t)$$



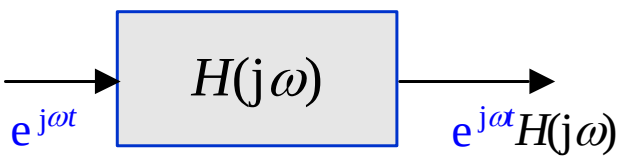


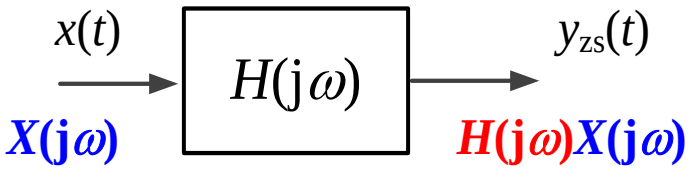
As the frequency increases, the amplitude response $|H(\mathrm{j}\omega)|$ of the system decreases continuously, indicating that the higher the frequency of the signal, the greater the loss of the signal passing through the system.

Because $|H(\mathrm{j}(1/RC))|\approx 0.7$, so $\omega_c=1/R$ is called the 3dB cut-off frequency of the system.

2. Frequency-domain analysis of non-periodic signals through continuous LTI systems

- Zero-state response of aperiodic signal through continuous LTI system





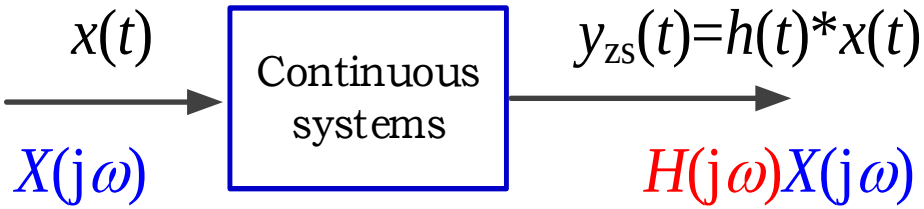
$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\mathrm{j}\omega)\,\underline{\mathrm{e}^{\mathrm{j}\omega t}}\,\mathrm{d}\omega$$

$$y_{zs}(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\mathrm{j}\omega)\underline{H(\mathrm{j}\omega)\mathrm{e}^{\mathrm{j}\omega t}}\,\mathrm{d}\omega$$

$$Y_{zs}(\mathrm{j}\omega) = H(\mathrm{j}\omega)X(\mathrm{j}\omega)$$

2. Frequency-domain analysis of non-periodic signals through continuous LTI systems

➢ **The relationship between the system zero state response** frequency domain analysis method **and** the convolution integration method**:**



The two analysis methods are essentially the same, except that the basic signal to express the signal is different.

The time-domain convolution theorem of Fourier transform is a bridge connecting the two.

2. Frequency-domain analysis of non-periodic signals through continuous LTI systems

Example It is known that the differential equation describing an LTI system is

$$y''(t) + 3y'(t) + 2y(t) = 3x'(t) + 4x(t),$$

Input stimulus $x(t) = e^{-3t}u(t)$, find the zero-state response $y_{zs}(t)$ of the system.

Solution: Since the spectral function of the input stimulus $x(t)$ is

$$X(j\omega) = \frac{1}{j\omega + 3}$$

The frequency response of the system can be obtained from the differential equation

$$H(j\omega) = \frac{3(j\omega) + 4}{(j\omega)^2 + 3(j\omega) + 2} = \frac{3(j\omega) + 4}{(j\omega + 1)(j\omega + 2)}$$

2. Frequency-domain analysis of non-periodic signals through continuous LTI systems

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Input stimulus $x(t) = e^{-3t}u(t)$ find the zero-state response $y_{zs}(t)$ of the system.

solution: The spectrum function $Y_{zs}(j\omega)$ of the zero-state response $y_{zs}(t)$ of the system is

$$Y_{zs}(j\omega) = X(j\omega)H(j\omega) = \frac{3(j\omega)+4}{(j\omega+1)(j\omega+2)(j\omega+3)}$$

$$= \frac{1/2}{j\omega+1} + \frac{2}{j\omega+2} - \frac{5/2}{j\omega+3}$$

$$y_{zs}(t) = \mathscr{F}^{-1}[Y_{zs}(j\omega)] = [\frac{1}{2}e^{-t} + 2e^{-2t} - \frac{5}{2}e^{-3t}]u(t)$$

2. Frequency-domain analysis of non-periodic signals through continuous LTI systems

Example It is known that the differential equation describing an LTI system is

$$y''(t) + 3y'(t) + 2y(t) = 3x'(t) + 4x(t),$$

Input stimulus(1)$x_1(t) = \cos(2t)u(t)$ (2)$x_2(t) = e^{2t}u(t)$,

Try to use the frequency domain method to analyze the zero state response of the system.

Analysis: (1) Due to $X_1(j\omega) = \frac{\pi}{2}[\delta(\omega-2)+\delta(\omega+2)] + \frac{j\omega}{4-\omega^2}$

so

$$Y_{1,zs}(j\omega) = X_1(j\omega)H(j\omega)$$

$$= \frac{\pi\delta(\omega-2)H(j2)}{2} + \frac{\pi\delta(\omega+2)H(-j2)}{2} + \frac{j\omega H(j\omega)}{4-\omega^2}$$

It is a difficult task to inverse Fourier transform the third term in the above formula.

2. Frequency-domain analysis of non-periodic signals through continuous LTI systems

Example It is known that the differential equation describing an LTI system is

$$y''(t) + 3y'(t) + 2y(t) = 3x'(t) + 4x(t),$$

Input stimulus $(1)x_1(t) = \cos(2t)u(t)$ $(2)x_2(t) = e^{2t}u(t)$,

Try to use the frequency domain method to analyze the zero state response of the system.

Solution: (2) Since the Fourier transform of $x_2(t) = e^{2t}u(t)$ **does not exist**,

Therefore, the zero-state response of the system cannot be solved by the frequency domain method.

3. The frequency domain response of a periodic signal passing through a continuous LTI system

- **Response of sinusoidal signal $\sin(\omega_0 t+\theta)$ through continuous LTI system**

From Euler's formula, it can be obtained

$$\sin(\omega_0 t+\theta) = (\mathrm{e}^{\mathrm{j}(\omega_0 t+\theta)} - \mathrm{e}^{-\mathrm{j}(\omega_0 t+\theta)})/2\,\mathrm{j}$$

Using the characteristics of the response of the imaginary exponential signal $\mathrm{e}^{\mathrm{j}\omega_0 t}$ and the linearity of the system, we can get

$$y_{\mathrm{zs}}(t) = \left[H(\mathrm{j}\omega_0)\mathrm{e}^{\mathrm{j}(\omega_0 t+\theta)} - H(-\mathrm{j}\omega_0)\mathrm{e}^{-\mathrm{j}(\omega_0 t+\theta)}\right]/2\mathrm{j}$$

$$|H(\mathrm{j}\omega_0)|\mathrm{e}^{\mathrm{j}\varphi(\omega_0)} \qquad |H(\mathrm{j}\omega_0)|\mathrm{e}^{-\mathrm{j}\varphi(\omega_0)}$$

$$= |H(\mathrm{j}\omega_0)|\sin(\omega_0 t + \varphi(\omega_0) + \theta)$$

3. The frequency domain response of a periodic signal passing through a continuous LTI system

- **Response of sinusoidal signal $\sin(\omega_0 t+\theta)$ through continuous LTI system**

$$\sin(\omega_0 t+\theta) \longrightarrow |H(\mathrm{j}\omega_0)|\sin(\omega_0 t+\varphi(\omega_0)+\theta)$$

The response of a sinusoidal signal with frequency ω_0 through a continuous LTI system is still a sinusoidal signal with the same frequency.

The magnitude change of the response is determined by $|H(\mathrm{j}\omega_0)|$,

The phase change of the response is determined by $\varphi(\omega_0)$.

in the same way

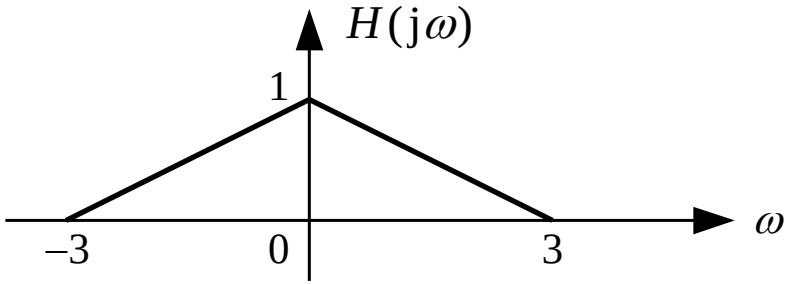
$$\cos(\omega_0 t+\theta) \longrightarrow |H(\mathrm{j}\omega_0)|\cos(\omega_0 t+\varphi(\omega_0)+\theta)$$

3. The frequency domain response of a periodic signal passing through a continuous LTI system

Example It is known that the frequency response of a continuous time system is shown in the figure, the input signal

$$x(t) = 5 + 3\cos 2t + \cos 4t \quad -\infty < t < \infty$$

Find the steady-state response $y(t)$ of the system。



solution:

Using the characteristics of the zero-state response of the cosine signal acting on the system, that is

$$T\{\cos(\omega_0 t + \theta)\} = \left|H(\mathrm{j}\omega_0)\right| \cos(\omega_0 t + \varphi(\omega_0) + \theta)$$

The steady-state response of the signal $x(t)$ acting on the system can be obtained as

$$T\{x(t)\} = 5H(\mathrm{j}0) + 3H(\mathrm{j}2)\cos 2t + H(\mathrm{j}4)\cos 4t$$

$$= 5 + \cos 2t \qquad -\infty < t < \infty$$

3. The frequency domain response of a periodic signal passing through a continuous LTI system

- Periodic signals pass through the zero-state response of the LTI system

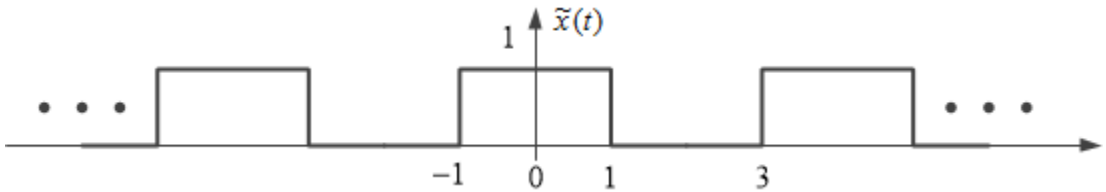
$$\tilde{x}(t) = \sum_n C_n \underline{e^{jn\omega_0 t}}$$

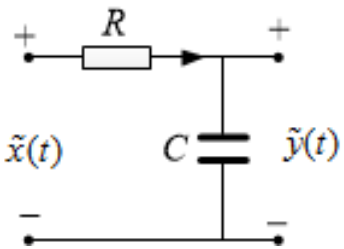
$$\Downarrow \qquad\qquad \Downarrow$$

$$\tilde{y}_{zs}(t) = \sum_n C_n \underline{H(jn\omega_0)e^{jn\omega_0 t}}$$

3. The frequency domain response of a periodic signal passing through a continuous LTI system

Example: Try to find the response of a rectangular wave with a period of 4 passing through the system as shown in the figure.





Solution: For a periodic rectangular signal, its Fourier coefficient is $C_n = 0.5\mathrm{Sa}\left(0.5\pi n\right)$

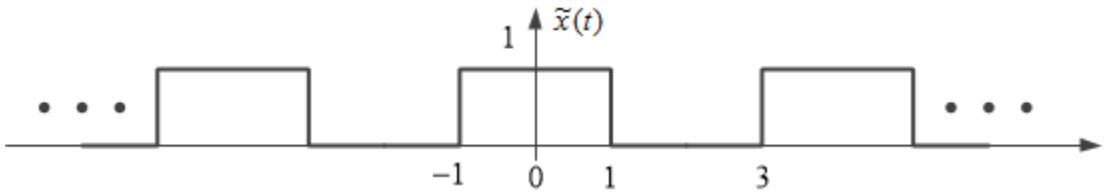
System Frequency Response $$H(\mathrm{j}\omega) = \frac{1/\mathrm{j}\omega C}{R + 1/\mathrm{j}\omega C} = \frac{1}{1 + \mathrm{j}\omega RC}$$

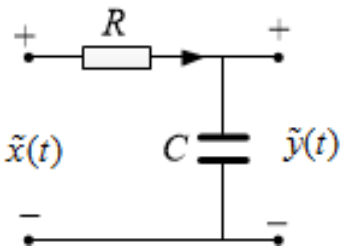
system output $$\tilde{y}(t) = \sum_{n=-\infty}^{\infty} C_n \cdot H(\mathrm{j}n\omega_0)\mathrm{e}^{\mathrm{j}n\omega_0 t}$$

$$= \sum_{n=-\infty}^{\infty} \mathrm{Sa}\left(0.5\pi n\right)\mathrm{e}^{\mathrm{j}n\pi t/2} / (1 + \mathrm{j}0.5n\pi RC)$$

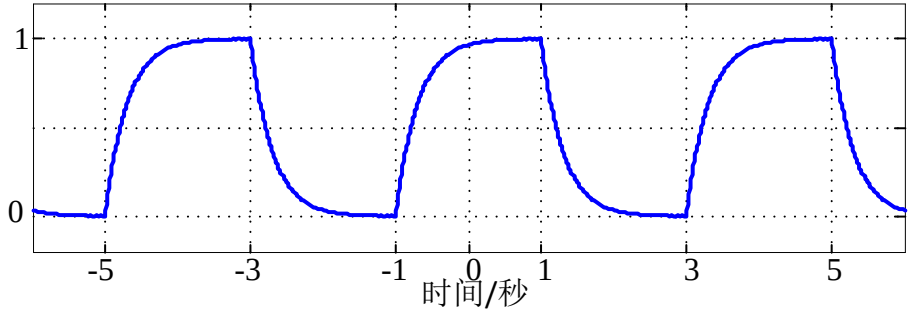
3. The frequency domain response of a periodic signal passing through a continuous LTI system

Example: Try to find the response of a rectangular wave with a period of 4 passing through the system as shown in the figure.

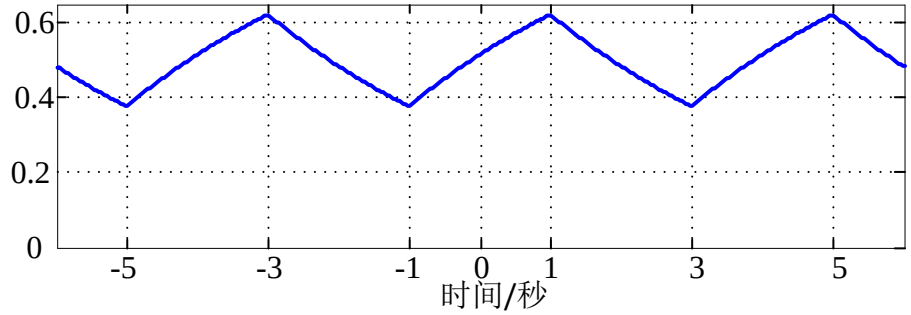




solution：



Output response of the system at RC=0.3 seconds



The output response of the system when RC=4 seconds

4. Distortion-free transmission system

Causes of distortion during signal transmission:

Non-linear distortion (creation of new frequency components)

Linear distortion (no new frequency components are generated)

Amplitude Distortion, Phase Distortion

There are two types of research on distortion problems in practical applications:

Signal transmission with as little distortion as possible (hi-fi system)

Deliberately distorting (predistorted waveform generation)

4. Distortion-free transmission system

If the input signal is $x(t)$, the output signal $y(t)$ of the distortion-free transmission system is

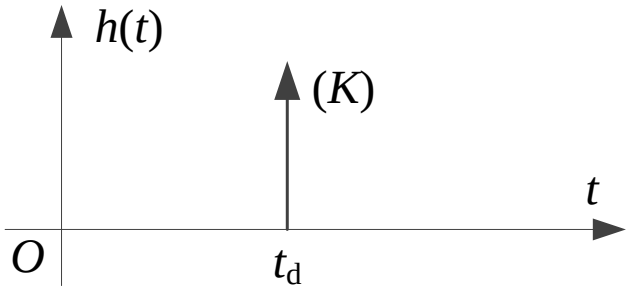
$$y(t) = K \cdot x(t - t_{\mathrm{d}})$$

K is a positive constant, and t_{d} is the delay time after the input signal passes through the system.

4. Distortion-free transmission system

- Time Domain Characteristics of Distortionless Transmission System

$$h(t) = K \cdot \delta(t - t_{\mathrm{d}})$$



Impulse Response of a Distortionless Transmission System

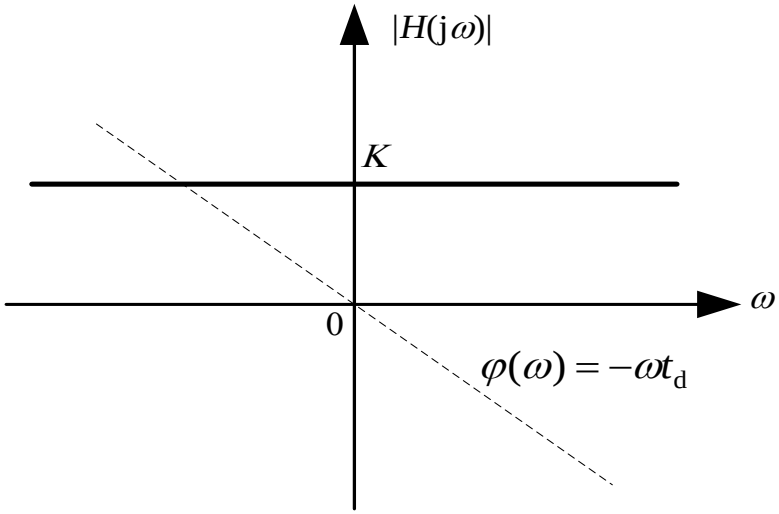
4. Distortion-free transmission system

- Magnitude and Phase Responses of Distortion-Free Transmission Systems

$$H(\mathrm{j}\omega) = K\mathrm{e}^{-\mathrm{j}\omega t_\mathrm{d}}$$

$$|H(\mathrm{j}\omega)| = K$$

$$\varphi(\omega) = -\omega t_\mathrm{d}$$



A distortion-free transmission system should satisfy two conditions:

※ The magnitude response of the system $|H(\mathrm{j}\omega)|$ is constant K over the entire frequency range, means that the bandwidth of the system is infinite;

※ The phase response of the system $\varphi(\omega)$ is linear with ω.

4. Distortion-free transmission system

Example It is known that the frequency response of a continuous LTI system is $H(\mathrm{j}\omega)=\dfrac{1-\mathrm{j}\omega}{1+\mathrm{j}\omega}$

(1) Calculate the magnitude response $|H(\mathrm{j}\omega)|$ and phase response $\varphi(\omega)$ of the system, And judge whether the system is a distortion-free transmission system.

(2) When the input is $x(t)=\sin t+\sin 3t$ $(-\infty<t<\infty)$, find the steady-state response of the system.

Solution: (1) Because $H(\mathrm{j}\omega)=1\cdot \mathrm{e}^{-\mathrm{j}2\arctan(\omega)}$

So the magnitude and phase responses of the system are

$$|H(\mathrm{j}\omega)|=1 \qquad \varphi(\omega)=-2\arctan(\omega)$$

The magnitude response of the system $|H(\mathrm{j}\omega)|$ is constant, but the phase response $\varphi(\omega)$ is not a linear function of ω, so the system is not a distortion-free transmission system.

4. Distortion-free transmission system

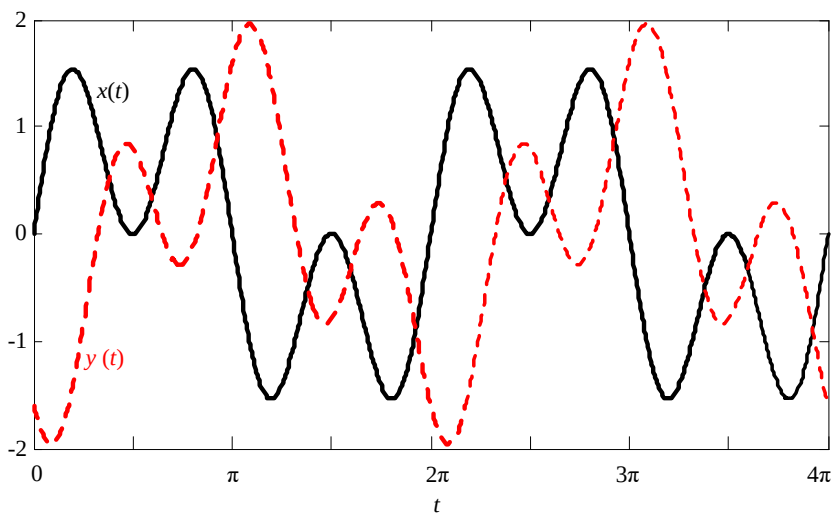
Example It is known that the frequency response of a continuous LTI system is $H(\mathrm{j}\omega)=\dfrac{1-\mathrm{j}\omega}{1+\mathrm{j}\omega}$

(1) Calculate the magnitude response $|H(\mathrm{j}\omega)|$ nd phase response $\varphi(\omega)$ of the system, And judge whether the system is a distortion-free transmission system.

(2) When the input is $x(t)=\sin t+\sin 3t$ $(-\infty<t<\infty)$, find the steady-state response of the system.

Solution: (2) $y(t)=|H(\mathrm{j}1)|\sin[t+\varphi(1)]+|H(\mathrm{j}3)|\sin[3t+\varphi(3)]$

$$=\sin(t-\pi/2)+\sin(3t-0.7952\pi)$$



Waveforms of input and output signals

output signal relative to The input signal is **distorted**.

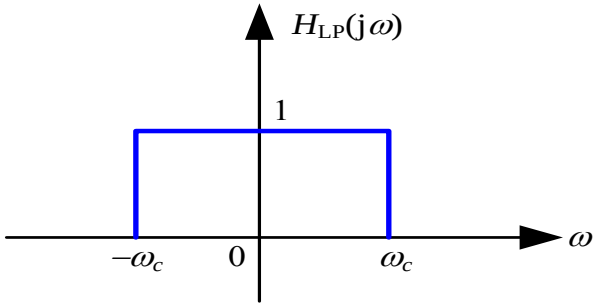
Distortion of the output signal is due to caused by the nonlinear phase of the system.

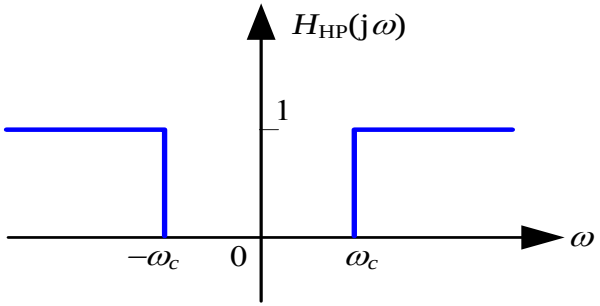
5. Ideal analog filter

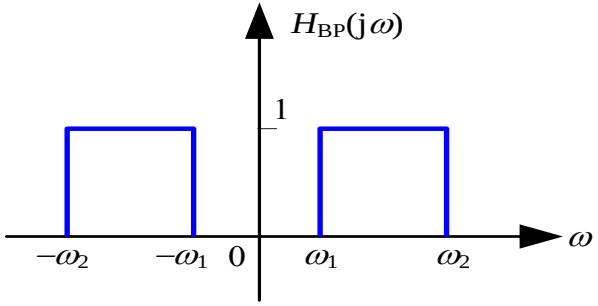
- Frequency Response of an Ideal Analog Filter
- Impulse Response of an Ideal Low-Pass Filter
- Step Response of an Ideal Low-Pass Filter

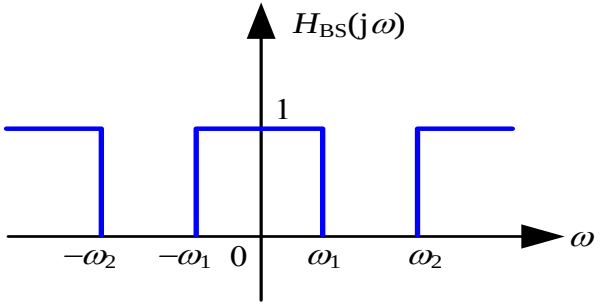
5. Ideal analog filter

- Frequency Response of an Ideal Analog Filter



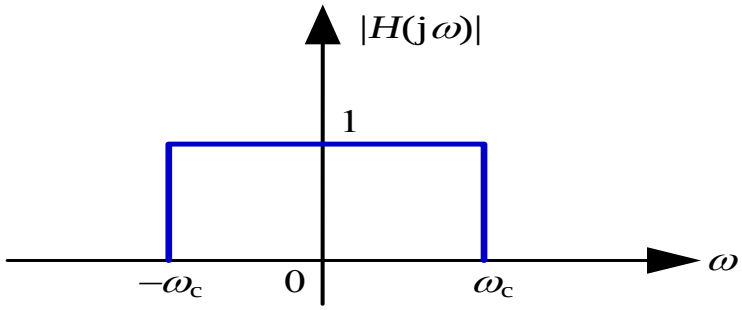


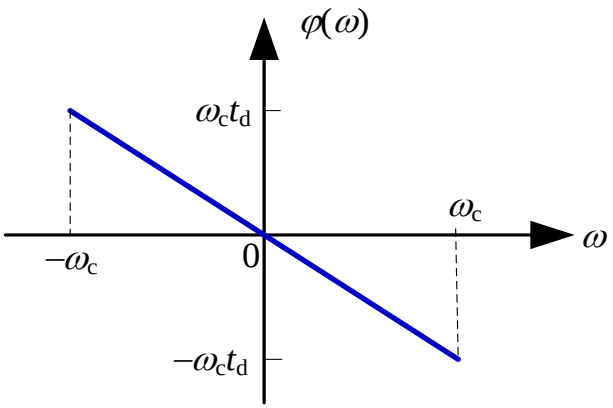




5. Ideal analog filter

- Frequency Response of Linear-Phase Ideal Low-Pass Filter



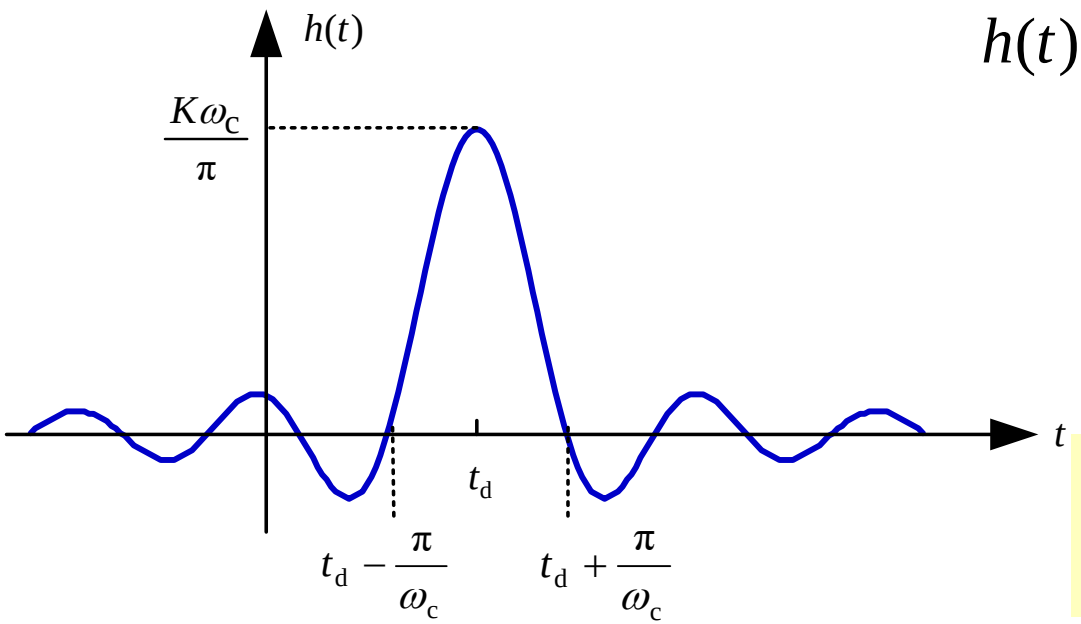


The amplitude response $|H(\mathrm{j}\omega)|$is always 1 in the passband($0\sim\omega_c$) and 0 in the stopband.

The phase response $\varphi(\omega)$ is linear with ω, $\varphi(\omega)= -\omega t_d$。

5. Ideal analog filter

- **Impulse Response** of Linear Phase Ideal Low-pass Filter



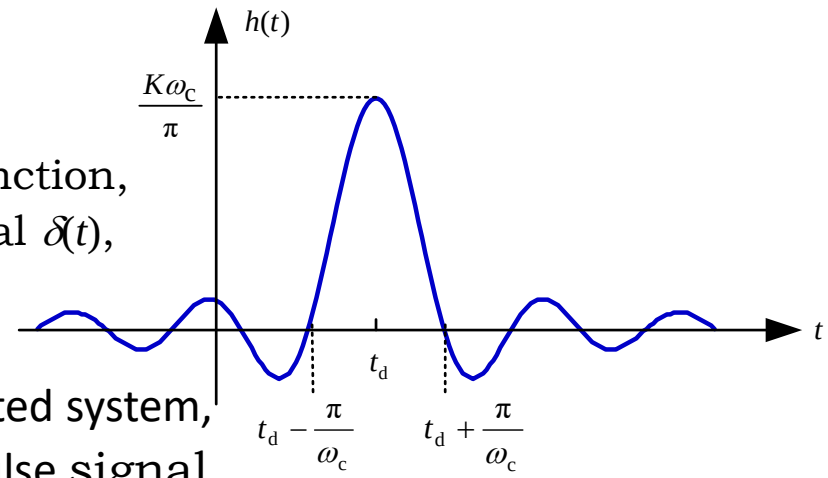
$$h(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} H(\mathrm{j}\omega)\mathrm{e}^{\mathrm{j}\omega t}\mathrm{d}t$$

$$= \frac{1}{2\pi}\int_{-\omega_c}^{\omega_c} \mathrm{e}^{-\mathrm{j}\omega t_d}\mathrm{e}^{\mathrm{j}\omega t}\mathrm{d}t$$

$$h(t) = \frac{\omega_c}{\pi}\mathrm{Sa}[\omega_c(t - t_d)]$$

5. Ideal analog filter

- Impulse Response of Linear Phase Ideal Low-pass Filter



※ The waveform of $h(t)$ is a sampling function, Unlike the waveform of the input signal $\delta(t)$, There is distortion.

Reason: The ideal low-pass filter is a band-limited system, And the frequency bandwidth of the impulse signal $\delta(t)$ is infinite.

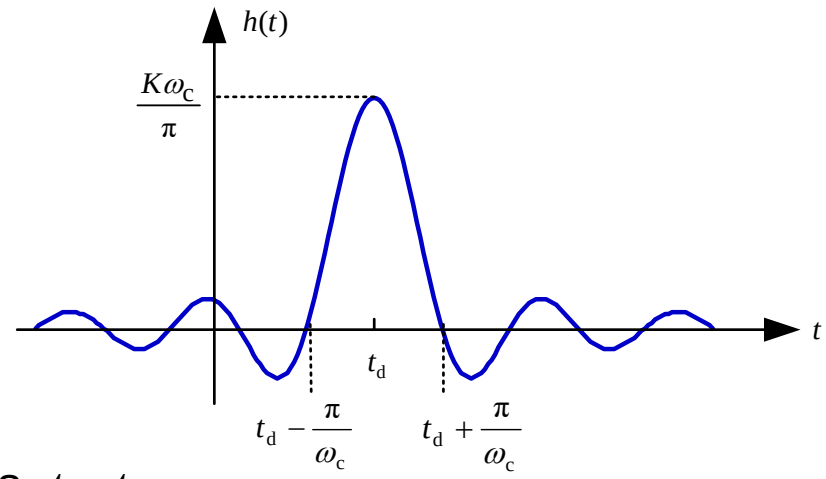
Distortion reduction method: increase the ideal low-pass cutoff ω_c.

When $\omega_c \to \infty$, the ideal low-pass becomes a distortion-free transmission system,

$h(t)$) also becomes an impulse signal.

5. Ideal analog filter

- **Impulse Response** of Linear Phase Ideal Low-pass Filter



※$h(t)$The time when the main peak appears $t = t_d$.

It is delayed by a time td from the moment$t = 0$ of the input signal $\delta(t)$.

$-t_d$ is the slope of the ideal low pass filter phase response.

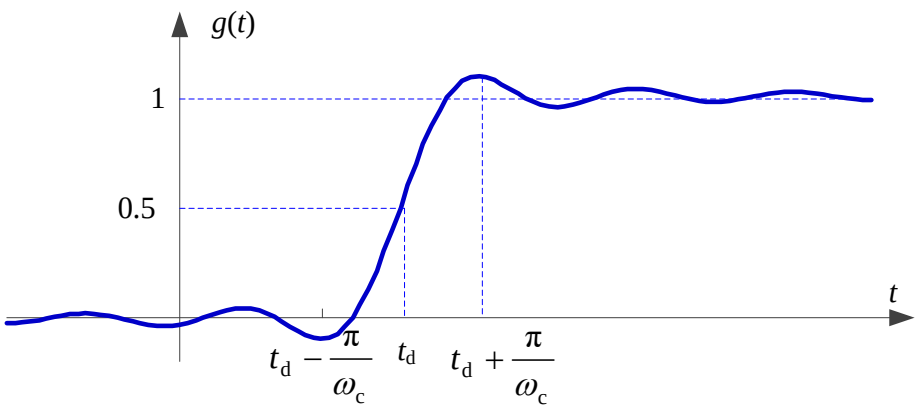
※ $h(t)$ also has an output in the interval of $t<0$, it can be seen that the ideal low-pass filter is a non-causal system, which is a physically unrealizable system.

5. Ideal analog filter

- **Step Response** of a Linear-Phase Ideal Low-Pass Filter

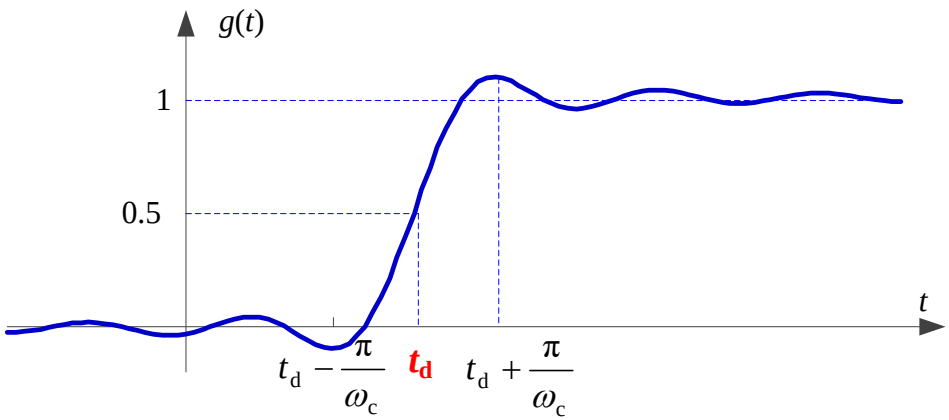
$$g(t)=h^{(-1)}(t)=\int_{-\infty}^{t}h(\tau)\mathrm{d}\tau \quad =\frac{\omega_{\mathrm{c}}}{\pi}\int_{-\infty}^{t}\mathrm{Sa}[\omega_{\mathrm{c}}(\tau-t_{\mathrm{d}})]\mathrm{d}\tau$$

$$g(t)=\frac{1}{2}+\frac{1}{\pi}\int_{0}^{\omega_{\mathrm{c}}(t-t_{\mathrm{d}})}\mathrm{Sa}(\lambda)\mathrm{d}\lambda$$



5. Ideal analog filter

- **Step Response** of a Linear-Phase Ideal Low-Pass Filter

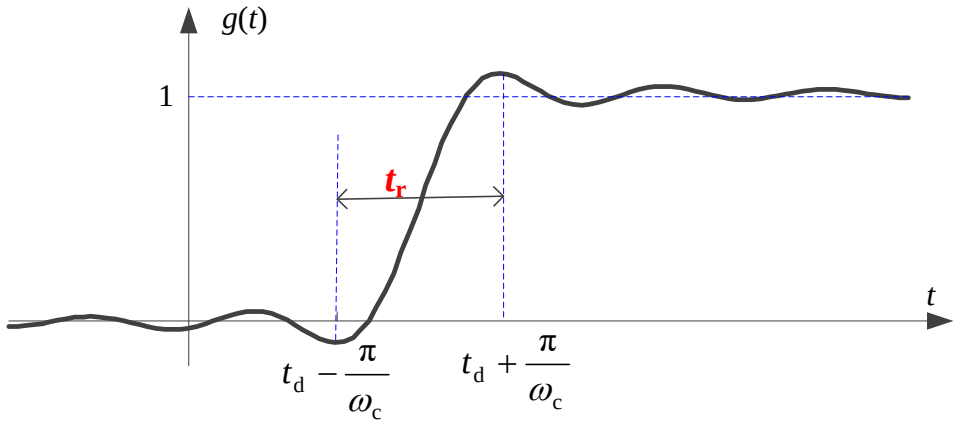


※ The step response $g(t)$ is delayed by t_d from the input step signal $u(t)$.

–t_d is the slope of the ideal low-pass filter phase response.

5. Ideal analog filter

- **Step Response** of a Linear-Phase Ideal Low-Pass Filter

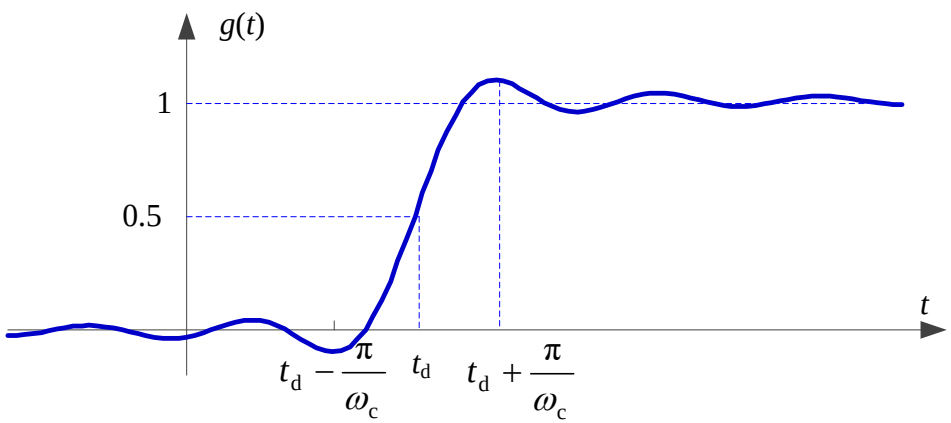


※ The rise time t_r of the step response $g(t)$ is inversely proportional to the ideal low-pass cutoff frequency ω_c.

The larger ω_c is, the shorter the rise time is, when $\omega_c \to \infty$, $t_r \to 0$.

5. Ideal analog filter

- **Step Response** of a Linear-Phase Ideal Low-Pass Filter



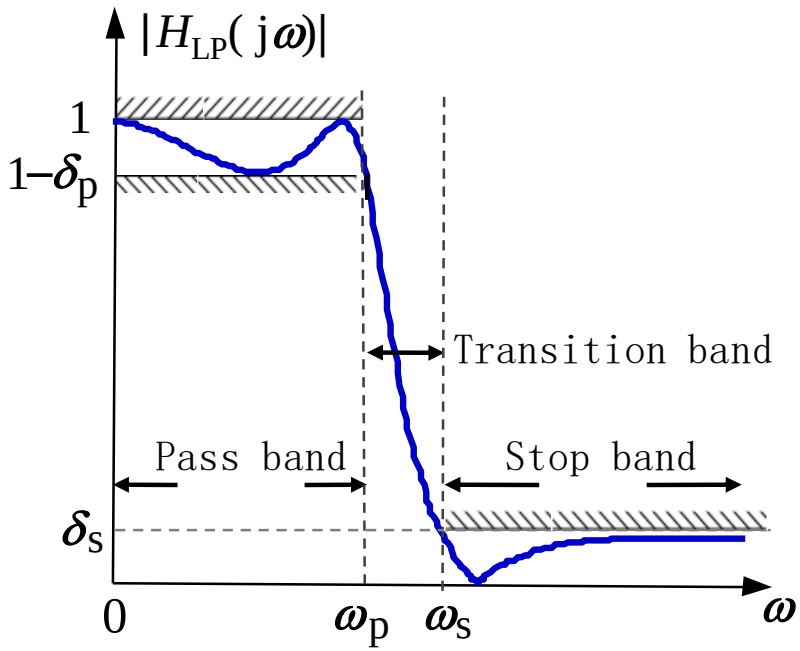
※ There is a Gibbs phenomenon: that is, oscillation appears near the discontinuity point, and the maximum peak value of the oscillation is about 9% of the step mutation value, and it does not decrease with the increase of the filter bandwidth.

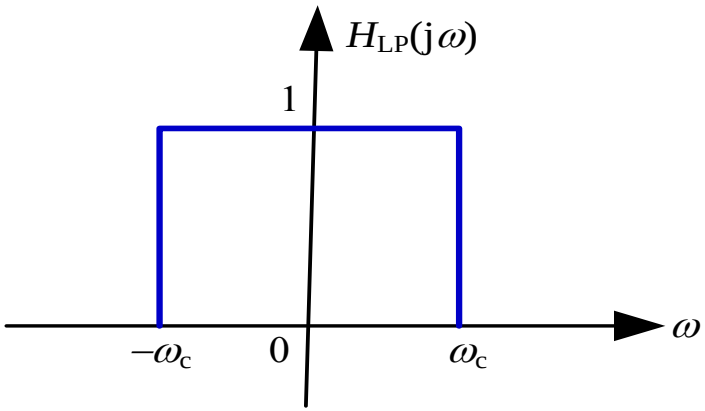
5. Ideal analog filter

1. The delay time of the output response depends on the slope of the phase response of the ideal low-pass filter.
2. After the input signal passes through the ideal low-pass filter, the output response produces a gradually rising or falling waveform at the discontinuity point of the input signal, and the rising or falling time is inversely proportional to the passband width of the ideal low-pass filter.
3. When the passband width of an ideal low-pass filter does not match the bandwidth of the input signal, the output will be distorted. If the passband width of the system is greater than the bandwidth of the signal, no transmission distortion will occur.

5. Ideal analog filter

※Actual low-pass analog filter



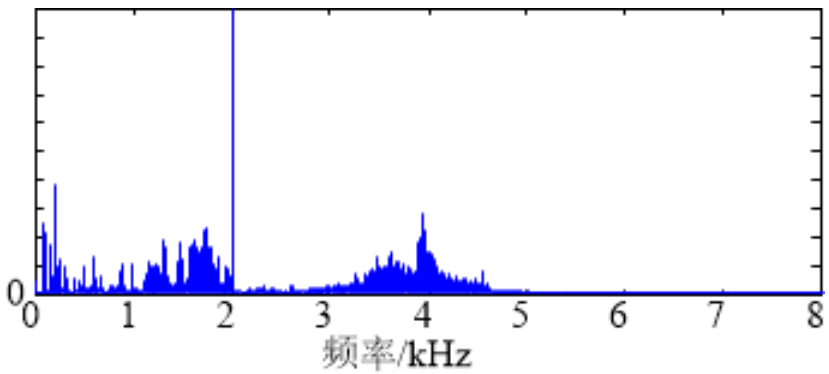


Ideal low pass filter

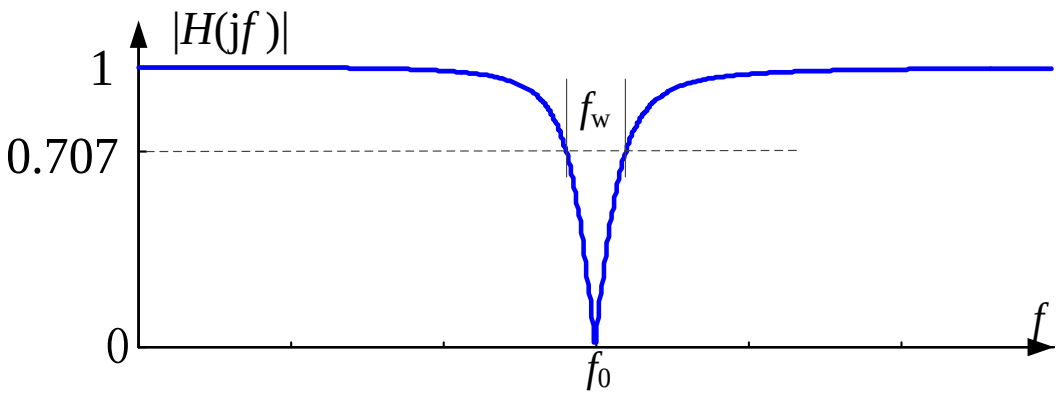
Example of Analog Filter Application

Example: The frequency spectrum of a music signal interfered by 2kHz noise is shown in the figure below.

Experiment with filters to remove the effects of noise.



disturbed by noise music signal spectrum

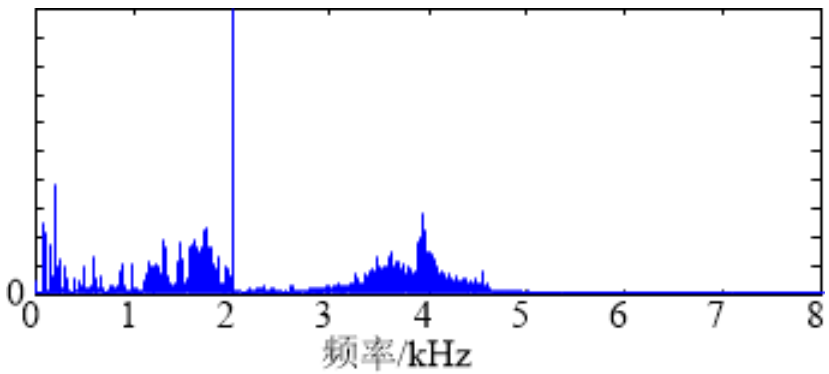


Magnitude Response of Second Order Bandstop Filter

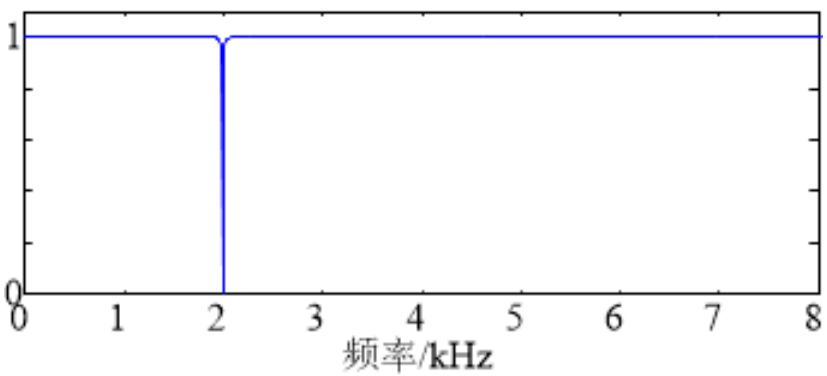
Example of Analog Filter Application

Example: The frequency spectrum of a music signal interfered by 2kHz noise is shown in the figure below.

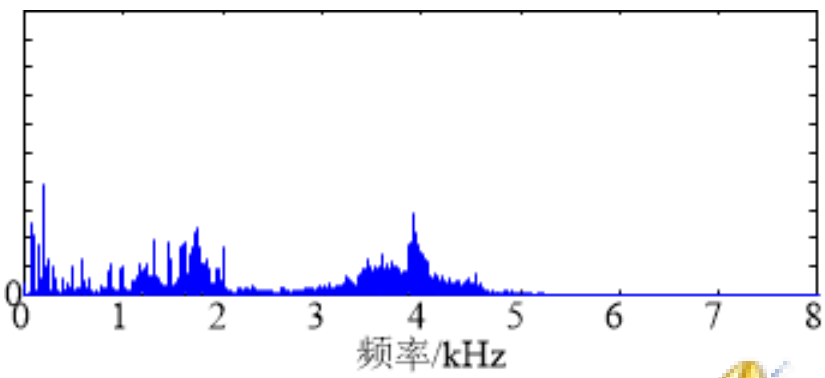
Experiment with filters to remove the effects of noise.



disturbed by noise music signal spectrum



f_0=2kHz, f_w=10Hz second-order band-stop filter magnitude response



Filtered music signal spectrum

Application example of frequency domain analysis for continuous time LTI system

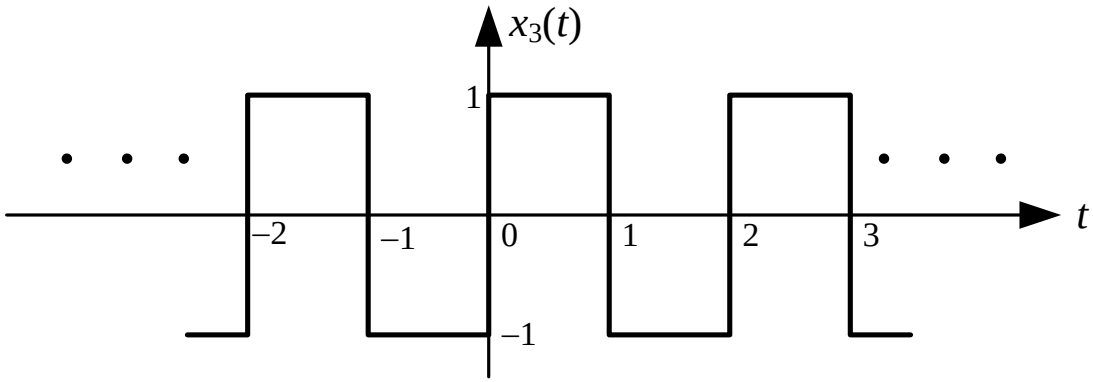
[Example] It is known that the frequency response $H(\mathrm{j}\omega)$ of the ideal analog low-pass filter is

$$H(\mathrm{j}\omega)=\begin{cases}\mathrm{e}^{-\mathrm{j}2\omega} & |\omega|<2\pi\\ 0 & |\omega|>2\pi\end{cases}$$

(1) If the system input signal $x_1(t)=\mathrm{Sa}(\pi t)$, try to find the system output response $y_1(t)$;

(2) If the system input signal $x_2(t)=\mathrm{Sa}(3\pi t)$, try to find the system output response $y_2(t)$;

(3) If the input signal $x_3(t)$ of the system is a periodic rectangular wave signal, try to find the output response $y_3(t)$ of the system.



Application example of frequency domain analysis for continuous time LTI system

[Example] It is known that the frequency response $H(\mathrm{j}\omega)$ of the ideal analog low-pass filter is

$$H(\mathrm{j}\omega)=\begin{cases}\mathrm{e}^{-\mathrm{j}2\omega} & |\omega|<2\pi\\ 0 & |\omega|>2\pi\end{cases}$$

Solution: (1) The signal $x_1(t)=\mathrm{Sa}(\pi t))$ passes through the response $y_1(t)$ of the system;

From the Fourier transform of the basic signal, we know that

$$\mathrm{Sa}(\pi t)\xrightarrow{\mathscr{F}}p_{2\pi}(\omega)$$

The output spectrum is

$$Y_1(\mathrm{j}\omega)=H(\mathrm{j}\omega)X_1(\mathrm{j}\omega)\ =p_{2\pi}(\omega)\mathrm{e}^{-\mathrm{j}2\omega}$$

It can be obtained by inverse Fourier transformation

$$y_1(t)=\mathscr{F}^{-1}\{Y_1(\mathrm{j}\omega)\}=\mathrm{Sa}\left[\pi(t-2)\right]$$

Application example of frequency domain analysis for continuous time LTI system

[Example] It is known that the frequency response $H(\mathrm{j}\omega)$ of the ideal analog low-pass filter is

$$H(\mathrm{j}\omega)=\begin{cases}\mathrm{e}^{-\mathrm{j}2\omega} & |\omega|<2\pi\\ 0 & |\omega|>2\pi\end{cases}$$

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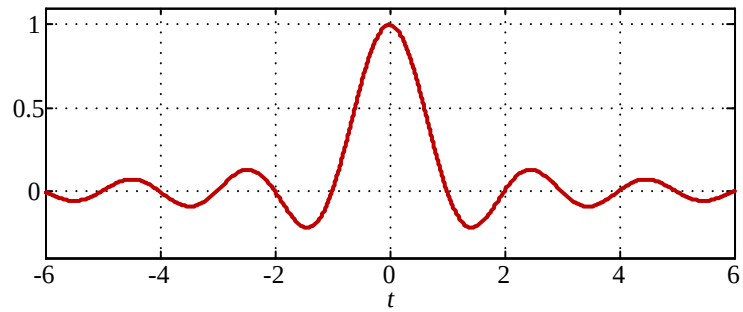
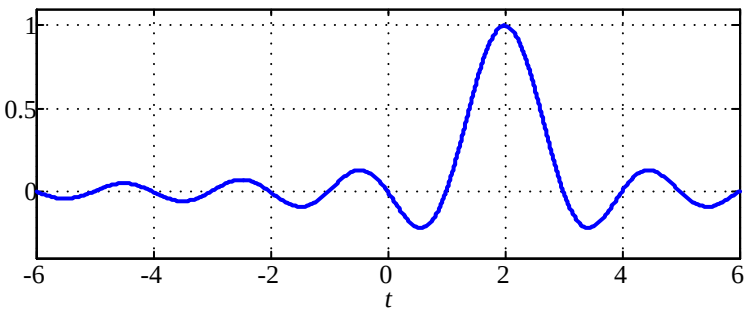


system input $x_1(t)=\mathrm{Sa}(\pi t)$



system output $y_1(t)=\mathrm{Sa}[\pi(t-2)]$

Application example of frequency domain analysis for continuous time LTI system

[Example] It is known that the frequency response $H(\mathrm{j}\omega)$ of the ideal analog low-pass filter is

$$H(\mathrm{j}\omega)=\begin{cases}\mathrm{e}^{-\mathrm{j}2\omega} & |\omega|<2\pi\\ 0 & |\omega|>2\pi\end{cases}$$

Solution: (2) The signal $x_2(t)=\mathrm{Sa}(3\pi t)$ passes through the response $y_2(t)$ of the system;

From the Fourier transform of the basic signal, we know that

$$\mathrm{Sa}(3\pi t)\xrightarrow{\mathscr{F}}(1/3)p_{6\pi}(\omega)$$

The output spectrum is

$$Y_2(\mathrm{j}\omega)=H(\mathrm{j}\omega)X_2(\mathrm{j}\omega)=(1/3)p_{4\pi}(\omega)\mathrm{e}^{-\mathrm{j}2\omega}$$

It can be obtained by inverse Fourier transformation $y_2(t)=\mathscr{F}^{-1}\{Y_2(\mathrm{j}\omega)\}=(2/3)\mathrm{Sa}\left[2\pi(t-2)\right]$

Application example of frequency domain analysis for continuous time LTI system

[Example] It is known that the frequency response $H(\mathrm{j}\omega)$ of the ideal analog low-pass filter is

$$H(\mathrm{j}\omega)=\begin{cases} \mathrm{e}^{-\mathrm{j}2\omega} & |\omega|<2\pi \\ 0 & |\omega|>2\pi \end{cases}$$

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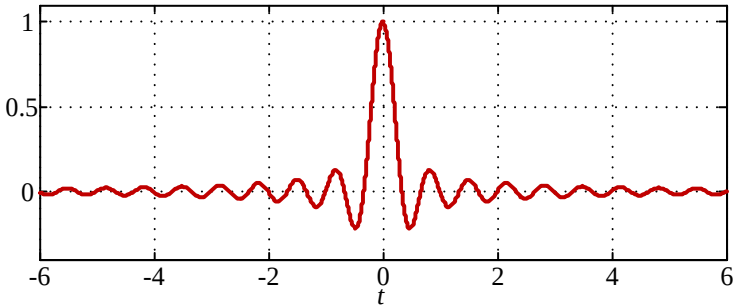
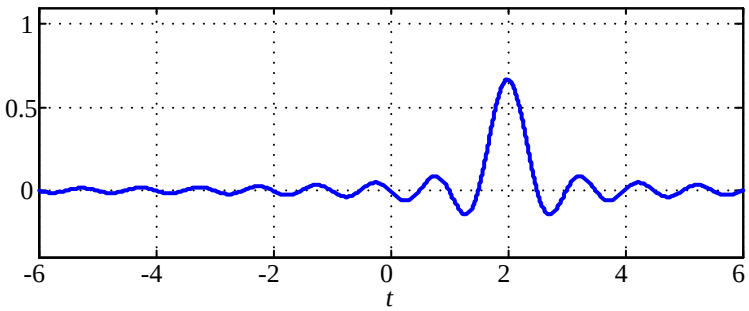


System input $x_2(t)=\mathrm{Sa}(3\pi t)$



system output$y_2(t)=(2/3)\mathrm{Sa}[2\pi(t-2)]$

Application example of frequency domain analysis for continuous time LTI system

[Example] It is known that the frequency response $H(\mathrm{j}\omega)$of the ideal analog low-pass filter is

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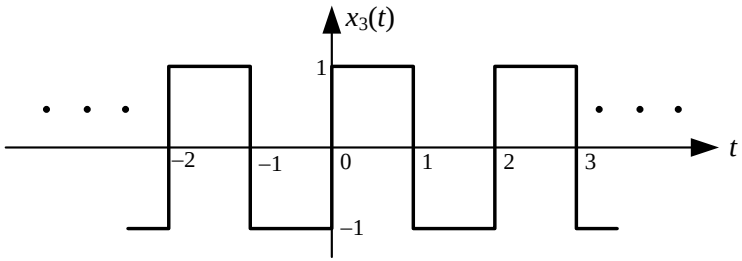

Solution: (3) Signal $x_3(t)$) passes through system response $y_3(t)$

fundamental frequency of the signal $\omega_0=2\pi/T_0=2\pi/2=\pi$

Since the cut-off frequency $\omega_c=2\pi$ of the ideal low-pass filter, only direct current and fundamental wave can pass through the ideal low-pass filter. Due to the real odd symmetry of $x_3(t)$, there is no DC component in $x_3(t)$.

$$C_1=\frac{1}{2}\int_0^2 x_3(t)\cdot\mathrm{e}^{-\mathrm{j}\pi t}\mathrm{d}t=\frac{2}{\mathrm{j}\pi}$$

$$C_{-1}=C_1^*=-\frac{2}{\mathrm{j}\pi}$$

Application example of frequency domain analysis for continuous time LTI system

[Example] It is known that the frequency response $H(\mathrm{j}\omega)$ of the ideal analog low-pass filter is

$$H(\mathrm{j}\omega)=\begin{cases}\mathrm{e}^{-\mathrm{j}2\omega} & |\omega|<2\pi\\ 0 & |\omega|>2\pi\end{cases}$$

Solution: (3) Signal $x_3(t)$passes through system response $y_3(t)$

$$x_3(t)=\sum_{n=-\infty}^{\infty}C_n\ \mathrm{e}^{\mathrm{j}n\pi t}$$

$$C_{-1}=-\frac{2}{\mathrm{j}\pi}\qquad C_0=0\qquad C_1=\frac{2}{\mathrm{j}\pi}$$

$$y_3(t)=C_{-1}H(-\mathrm{j}\pi)\mathrm{e}^{-\mathrm{j}\pi(t-2)}+C_1H(\mathrm{j}\pi)\mathrm{e}^{\mathrm{j}\pi(t-2)}=\frac{4}{\pi}\sin(\pi t)$$

Application example of frequency domain analysis for continuous time LTI system

[Example] It is known that the frequency response $H(\mathrm{j}\omega)$ of the ideal analog low-pass filter is

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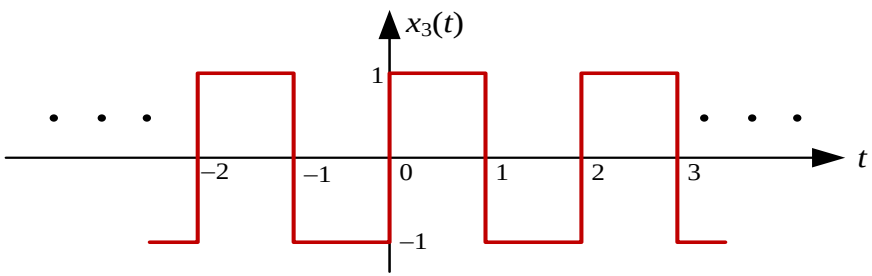
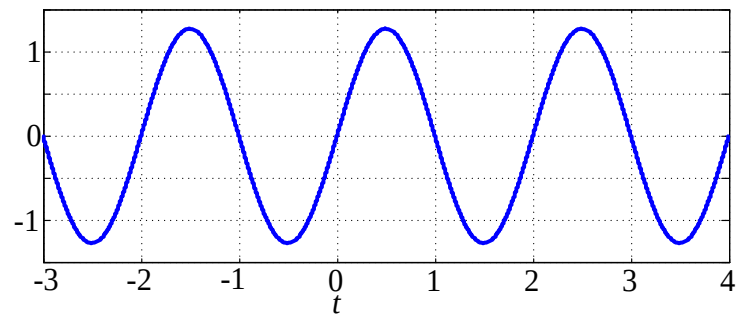


System input $x_3(t)$



system output $y_3(t)=4\sin(\pi t)/\pi$

Frequency Domain Analysis of Continuous Time LTI System Response

- Advantages: When solving the zero-state response of the system, it can intuitively reflect the change of the signal spectrum after the signal passes through the system, explain the difference between the excitation and response time-domain waveforms, and the physical concept is clear.
- insufficient:

(1) Only the zero-state response of the system can be solved, and the zero-input response of the system

It still needs to be solved by the time domain method.

(2) If the Fourier transform of the system frequency response (or excitation signal) does not exist,Then the frequency domain analysis method cannot be used.Then the frequency domain analysis method cannot be used.

- Solution: use Laplace transform

Thanks

Some of the materials cited in this course are the teaching accumulation of the main lecturer for many years, which come from a variety of media and the exchanges of colleagues, peers and friends, and it is difficult to indicate the source one by one, so I hereby explain and thank you!